

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			B1	ALL BOLTS ARE METRIC DIA.
20										B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
20			INDUC BEND	9001818CTD	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
30			INDUC BEND	9001813	6	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
40			FLANGE	9002644	0.75	2	NIPOFLANGE 300# RF 7.14 MM/XS WT LTCS A694 F60	12	12	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
50	U04	VAV	MODULAR DBB	K03AC02	0.75	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
60	U04		MODULAR DBB	M03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
70	U04	VAW	MODULAR DBB	M03EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2		
80	U04	VAW	MODULAR DBB	M03EC02	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3		
90							***** BALL VALVES *****				
90											
100											
100	U04		BALL	B03AC01	0.75	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
110	U04	VA5	BALL	B03AC02	0.75	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
120	U04		BALL	B03AC03	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0		
130	U04		BALL	B03AC07	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
140	U04		BALL	B03JC07	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A	0	0		
150							***** GATE VALVES *****				
150											
160	U04		GATE	W03AC70	3	6	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
170	U04		GATE	W03AC71	8	24	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
180			GATE	W03JC71	28	28	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A	0	0		
190							***** PIPE ABOVE GROUND*****				
190											
200											
200			PIPE	5358508	0.75	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0		
210			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
220			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		
230											
230			PIPE	9001257	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0		
240			PIPE	9001254	6	10	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE	0	0		
250			PIPE	9001256	12	12	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE	0	0		
260			PIPE	9001311	14	16	LINEPIPE SMLS WT=7.92MM API-5L X60 BE	0	0		
270			PIPE	9001485	18	18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE	0	0		
280			PIPE	9001493	20	20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE	0	0		
290			PIPE	9001494	24	24	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE	0	0		
300			PIPE	9001490	28	28	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE	0	0		
310							*****PIPE BELOW GROUND*****				
310											
320											
320			PIPE	9001257CTD	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED	0	0		
330			PIPE	9001254CTD	6	10	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0		
340			PIPE	9001256CTD	12	12	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE COATED	0	0		
350			PIPE	9001311CTD	14	16	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED	0	0		
360			PIPE	9001487CTD	18	18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED	0	0		
370			PIPE	9001493CTD	20	20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0		
380			PIPE	9001493CTD	24	24	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0		
390			PIPE	9001490CTD	28	28	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE COATED	0	0		
400							*****FITTINGS ABOVE AND BELOW GROUND*****				
400											
410											
410			ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
420			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0		
430	U04		ELBOW	9006599	24	24	ELL BW 90 DEG LR WT 10.31 MM A860 WPHY 60	0	0		
440	U04		ELBOW	9006478	16	16	ELL BW 90 DEG LR WT 7.92 MM A860 WPHY 60	0	0		
450			ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
460			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0		
470											
470			TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0		
480			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0		
490			TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2		
500			TEE RED	5392454	0.75	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3		
510			TEE RED	9006017	6	6	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	16	16		
520			TEE RED	9004149	8	8	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W	20	20		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C	12/16/2024	BY OZAC	U04	Page 1 of 4	300K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
530											
530			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0		
540			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0		
550			CAP	9005062	4	4	CAP BW WT 6.02 MM LTCS A860 WPHY-60	0	0		
560			CAP	9005055	6	10	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0		
570			CAP	9005057	12	12	CAP BW WT 7.14 MM LTCS A860 WPHY-60	0	0		
580			CAP	9005058	14	16	CAP BW WT 7.92 MM LTCS A860 WPHY-60	0	0		
590			CAP	9005059	18	18	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0		
600			CAP	9005060	20	20	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0		
610			CAP	9005064	24	24	CAP BW WT 11.13 MM LTCS A860 WPHY-60	0	0		
620			CAP	9005065	28	28	CAP BW WT 11.91 MM LTCS A860 WPHY-60	0	0		
630	U04		CAP	9005061	24	24	CAP BW WT 10.31 MM LTCS A860 WPHY-60	0	0		
640											
640			RED CONC	5392458	0.75	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1	2		
650			RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2		
660			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3		
670			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3		
680			RED ECC	9001372	1.5	2	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4		
690			RED ECC	9002281	3	3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4	4		
700			RED ECC	9002012	3	3	REDUCER ECC WT 6.35 MM / 5.49 MM LTCS A860 WPHY-60	6	6		
710			RED ECC	9002312	4	4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6	10		
720			RED ECC	9002266	6	8	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8	10		
730			RED ECC	9002269	6	10	REDUCER ECC WT 7.14 MM / 6.35 MM LTCS A860 WPHY-60	12	12		
740			RED ECC	9002272	6	10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	14	14		
750			RED ECC	9002272	8	10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	16	16		
760			RED ECC	9002278	14	14	REDUCER ECC WT 7.92MM LTCS A860 WPHY-60	16	16		
770			RED ECC	9002292	10	10	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	18	18		
780			RED ECC	9002294	12	12	REDUCER ECC WT 8.74 MM / 7.14 LTCS A860 WPHY-60	18	18		
790			RED ECC	9002250	14	16	REDUCER ECC WT 8.74 MM / 7.92 MM LTCS A860 WPHY-60	18	18		
800	U04		RED ECC	9004134	12	12	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60-W	20	20		
810	U04		RED ECC	9004135	14	16	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	20	20		
820	U04		RED ECC	9004136	18	18	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	20	20		
830	U04		RED ECC	9004144	16	16	REDUCER ECC WT 10.31 MM / 7.92 MM LTCS A860 WPHY-60-W	24	24		
840			RED ECC	9004145	18	18	REDUCER ECC WT 10.31 MM / 8.74 MM LTCS A860 WPHY-60-W	24	24		
850	U04		RED ECC	9004145	18	18	REDUCER ECC WT 10.31 MM / 8.74 MM LTCS A860 WPHY-60-W	24	24		
860	U04		RED ECC	9004137	20	20	REDUCER ECC WT 10.31 MM / 9.53 MM LTCS A860 WPHY-60-W	24	24		
870			RED ECC	9004146	18	18	REDUCER ECC WT 11.91MM / 8.74MM LTCS A860 WPHY-60-W	28	28		
880			RED ECC	9004147	20	20	REDUCER ECC WT 11.91 MM / 9.53 LTCS A860 WPHY-60-W	28	28		
890			RED ECC	9004148	24	24	REDUCER ECC WT 11.91 MM / 10.31 LTCS A860 WPHY-60-W	28	28		
900											
900							***** INDUCTION BENDS ABOVE GROUND*****				
910											
910			INDUC BEND	9001810	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
920			INDUC BEND	9001814	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
930			INDUC BEND	9001815	14	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
940			INDUC BEND	9001816	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
950			INDUC BEND	9001817	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
960			INDUC BEND	9001818	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
970			INDUC BEND	9001397	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
980			INDUC BEND	9001820	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
990			INDUC BEND	9001823	6	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1000			INDUC BEND	9001824	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1010			INDUC BEND	9001825	14	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1020			INDUC BEND	9001826	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1030			INDUC BEND	9001827	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1040	U04		INDUC BEND	9001828	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1050			INDUC BEND	9001828	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1060			INDUC BEND	9001500	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1070			INDUC BEND	9001830	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1080			INDUC BEND	9001833	6	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1090			INDUC BEND	9001834	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1100			INDUC BEND	9001835	14	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1110			INDUC BEND	9001836	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1120			INDUC BEND	9001837	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1130			INDUC BEND	9001838	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1140			INDUC BEND	9002001	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L	0	0		
1150											
1150											
1160							***** INDUCTION BENDS BELOW GROUND*****				
1160			INDUC BEND	9001810CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1170			INDUC BEND	9001813CTD	6	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
1180			INDUC BEND	9001814CTD	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BWCOATED	0	0		

	TENGIZCHEVROI TCO TENGI, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: - 40 / 80 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	300K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1190			INDUC BEND	9001815CTD	14	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1200			INDUC BEND	9001816CTD	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1210			INDUC BEND	9001817CTD	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1220			INDUC BEND	9001397CTD	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1230			INDUC BEND	9001820CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1240			INDUC BEND	9001823CTD	6	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1250			INDUC BEND	9001824CTD	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1260			INDUC BEND	9001825CTD	14	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1270			INDUC BEND	9001826CTD	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1280			INDUC BEND	9001827CTD	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1290	U04		INDUC BEND	9001828CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1300			INDUC BEND	9001828CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1310			INDUC BEND	9001500CTD	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1320	U04		INDUC BEND	9007008CTD	24	24	INDUCTION BEND 45DEG X 10D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1330	U04		INDUC BEND	9007023CTD	24	24	INDUCTION BEND 45DEG X 15D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1340			INDUC BEND	9001830CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1350			INDUC BEND	9001833CTD	6	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1360			INDUC BEND	9001834CTD	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1370			INDUC BEND	9001835CTD	14	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1380			INDUC BEND	9001836CTD	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1390			INDUC BEND	9001837CTD	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1400			INDUC BEND	9001838CTD	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1410			INDUC BEND	9002001CTD	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L	0	0	
1420							*****OLETS ABOVE & BELOW GROUND WT*****			
1420										
1430										
1430	U04		OLETS	5505333	1	1	WELDOLET SCH XS LTCS A350 LF2 CL.1	2	2	
1440			OLETS	9006407	0.75	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	10	
1450			OLETS	9006409	3	3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8	10	
1460			OLETS	9006413	0.75	2	WELDOLET WT 7.14 MM/XS WT LTCS A694 F60	12	12	
1470			OLETS	9006412	3	4	WELDOLET WT 7.14 MM/STD WT LTCS A694 F60	12	12	
1480			OLETS	9002552	0.75	2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	14	16	
1490			OLETS	9006415	3	4	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	14	16	
1500			OLETS	9006423	0.75	2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	18	18	
1510			OLETS	9006422	3	4	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	18	18	
1520			OLETS	9002008	6	6	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	18	18	
1530			OLETS	9006429	0.75	2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	20	20	
1540			OLETS	9006428	3	4	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	20	20	
1550			OLETS	9006424	6	6	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	20	20	
1560			OLETS	9002551	0.75	2	WELDOLET WT 10.31MM/ XS WT LTCS A694 F60	24	24	
1570			OLETS	9006434	3	4	WELDOLET WT 10.31 MM/STD WT LTCS A694 F60	24	24	
1580			OLETS	9006430	6	8	WELDOLET WT 10.31 MM/6.35 MM LTCS A694 F60	24	24	
1590			OLETS	9002003	0.75	2	WELDOLET WT 11.91MM/SCH XS LTCS A694 F60	28	28	
1600			OLETS	9002014	3	3	WELDOLET WT 11.91MM/5.49MM LTCS A694 F60	28	28	
1610			OLETS	9002004	4	4	WELDOLET WT 11.91MM/6.02MM LTCS A694 F60	28	28	
1620			OLETS	9002009	6	10	WELDOLET WT 11.91MM/6.35MM LTCS A694 F60	28	28	
1630										
1630	U04		FLANGE	5504021	0.75	2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	3	
1640			FLANGE	9002639	0.75	2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4	4	
1650			FLANGE	9002643	0.75	2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6	10	
1660			FLANGE	9002645	0.75	2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	14	16	
1670			FLANGE	9002646	0.75	2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	18	18	
1680			FLANGE	9002647	0.75	2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	20	20	
1690			FLANGE	9002010	0.75	2	NIPOFLANGE 300# RF 11.91 MM/XS WT LTCS A694 F60	28	28	
1700										
1700							***** FLANGES ABOVE GROUND*****			
1710										
1710	U04	FA8	FLANGE	5599703	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	1.5	
1720	U04		FLANGE	5556476	0.5	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1730	U04		FLANGE	5556473	3	3	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
1740			FLANGE	9001480	4	4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	300K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1750			FLANGE	9001483	6	12	FLG WN 300# RF LTCS A694 F60 WT 6.35MM	0	0	
1760			FLANGE	9001484	12	12	FLG WN 300# RF LTCS A694 F60 WT 7.14MM	0	0	
1770			FLANGE	9001484	12	16	FLG WN 300# RF LTCS A694 F60 WT 7.14MM	0	0	
1780			FLANGE	9001455	14	18	FLG WN 300# RF LTCS A694 F60 WT 7.92MM	0	0	
1790			FLANGE	9001457	18	20	FLG WN 300# RF LTCS A694 F60 WT 8.74MM	0	0	
1800			FLANGE	9001458	20	24	FLG WN 300# RF LTCS A694 F60 WT 9.53MM	0	0	
1810			FLANGE	9001460	24	24	FLG WN 300# RF LTCS A694 F60 WT 10.31MM	0	0	
1820			FLANGE	9002011	28	28	FLG WN 300# RF LTCS A694 F60 WT 11.91MM SERIES A	0	0	
1830	U04		BLIND	5615576	0.75	3	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
1840			BLIND	9001450	4	24	FLG BLIND 300# RF LTCS A694 F60	0	0	
1850			BLIND	9001451	28	28	FLG BLIND 300# RF LTCS A694 F60 SERIES A	0	0	
1860							***** FLANGES BELOW GROUND*****			
1870										
1870			FLANGE	9001459	28	28	FLG WN 300# RF LTCS A694 F60 WT 10.31MM SERIES A	0	0	
1880							***** GASKETS *****			
1890										
1890			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1900			GASKET	5674032	28	28	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
1910										
1910		B1					***** BOLTING *****			
1920										
1920	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1930										
1930		SB6	BOLTING	5675675	12	28	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1940										
1940			SPEC BLIND	5749581	0.75	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
1950										
1950			PADDLE	5750100	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
1960			PADDLE	5750028	28	28	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
1970			PADDLE	5750200	12	24	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
1980			PADDLE	5750198	28	28	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
1990										

	TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	300K5F